



AI's tightest bottleneck: Memory chips

June 18, 2026

Figure 1: Key Terminology

DRAM	Dynamic Random-Access Memory: One of two main memory technologies (50-60% of annual industry revenue). DRAM is fast, volatile working memory that serves as a temporary system memory. It offers low latency for frequent reads/writes, but is volatile and must be refreshed constantly, which adds to power use. It primarily sells into data centre, client PC, graphics, industrial and automotive end-markets, and is highly concentrated, dominated by three players: SK Hynix (36%), Samsung (33%), and Micron (24%).
NAND	NOT-AND: One of two main memory technologies (30-50% of annual industry revenue). NAND is non-volatile flash storage used to keep data when power is off. Because of this it is useful for applications in the data centre, client PC, consumer and automotive end markets.
HBM	High Bandwidth Memory: A specialized form of DRAM, built by stacking individual DRAM dies vertically to create the ultra-fast, high-bandwidth memory used directly in AI accelerators and high-performance computing. Key applications of HBM include AI/Machine Learning, Data Centres, Graphics Processing and High-Performance Computing.

Source: Deutsche Bank Research

In 2026, we can no longer ignore the memory market's parabolic rise. The largest memory chipmakers, SK Hynix, Micron and Samsung Electronics are each now valued above \$1trn in market cap – a feat considering they represent over 90% of the DRAM market. What is happening in the memory market to drive these geometric valuations? Behind the ascent, the answer is more worrying than current prices would seem to suggest.

The reason is that the global race for AI is driving a supply-displacement shock into the memory market. And as the industry shifts from standard large language model (LLM) response to autonomous agents that observe, reason, plan and act (Agentic AI), this problem will only get more acute.

While memory has historically suffered a boom-and-bust cycle, the exponential demand from AI suggests the current cycle is more structural than cyclical. In this case, chipmakers' ability to deliver cannot keep pace with the surge in hyperscale demand for memory used in AI. And given it takes years for chipmakers to build fabs for memory chip production, we should expect a multi-year supply shock that constrains the rate of progress in AI. More worryingly for corporations and consumers, the vast amount of memory being swallowed to produce HBMs means less is being allotted to produce other types such as standard DRAM, which is used in the everyday applications the global economy depends on.

As a result, the shortage facing the memory industry has massive repercussions for the individual consumer. Just this month, nine US trade associations representing the automotive, consumer electronics, medical devices, telecoms and retail industries sent a letter to US Treasury Secretary Scott Bessent and Commerce Secretary Howard Lutnick warning of the consequences the AI-fuelled rush for memory will have on the US economy.¹

Lastly, the memory shortage creates a new dimension in US-China tech competition. For China, the repercussions are double-edged. On the one hand, the shortage constrains

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¹ <https://cryptobriefing.com/us-memory-chip-shortage-ai-boom/>



Chinese chipmakers further in the broader logic-chip and AI build-out. China's AI self-sufficiency is constrained by its access to both high-end compute and memory. Without them, its performance in AI processing and inference remains limited. On the other hand, the vacuum of memory chips needed to power traditional devices opens a space for lagging-edge (older, less advanced process node) Chinese memory makers like ChangXin Memory Technologies and Yangtze Memory Technologies Corporation. As Western non-AI corporations are squeezed out of the memory market, the possibility they could turn to Chinese memory manufacturers instead is growing.

1. The squeeze: What's actually happening in memory?

(i) The demand side: AI's bottomless appetite

The problem facing the memory industry is that there is not enough supply to meet the exponential demand for AI inference.

The crisis partially lies in high-bandwidth memory (HBM), a "memory stack" that layers individual DRAM dies into the ultra-fast, 3D memory needed to process AI models. For example, an AI chip, such as an Nvidia GPU, can only compute on data that is loaded onto it; memory does the holding and feeding of that data into the models. Memory has two dimensions: capacity (how much data fits) and bandwidth (how fast it moves). Without memory, chips cannot compute or train AI models – and without it, companies cannot run a trained model to perform an agentic task (inference).

In addition, the fast-evolving AI landscape is beginning to require more expansive memory consumption across other areas of the memory stack beyond. In particular, the shift from human-driven generative AI to inference and agent-driven AI is driving another leg up for DRAM bit demand. Agentic AI systems, unlike traditional AI models, can store and recall past experiences, learn from interactions, and maintain context, leading to more coherent outputs. This requires various types of memory to function effectively, spanning across other memory devices (DDR5, LPDDR, NAND).

Memory also dictates the pace of AI advancement, because companies cannot scale on compute alone. This is the "memory wall" problem: beyond a point, peak computational performance cannot be reached without expanding memory bandwidth. In other words, increasing computing power without increasing memory bandwidth eventually limits the benefits of the compute itself.

Thus, the accelerating need for memory is – alongside compute – a defining bottleneck for AI progress². But memory has arguably become a tighter near-term constraint, given its manufacturing and supply limits: HBM bandwidth for inference-heavy workloads, HBM/packaging capacity, and capacity for traditional DRAM for long-context queries as the determining factor for how fast compute can be deployed. Micron's CEO has called memory the backbone of AI, noting: "as AI models grow larger and inference requirements continue to rise – AI moving from training to inference, and from data centers to the edge – the demand for storage will only increase. It requires greater capacity, higher performance, and lower power consumption."³ As a result, hyperscalers (the largest cloud operators) like Meta, Amazon and Microsoft need vast quantities of HBM both to hold the model and to keep the compute from sitting "idle." In recent years, they have driven the memory market and last year total revenues increased by 35% y/y to a record \$223bn⁴.

² In the AI tech stack, compute is made of GPUs, TPUs, and other AI accelerators, while memory refers to where the data sits. That can include the model's weights, the intermediate activations produced during a forward pass, the KV cache used during inference, etc. Memory has two relevant dimensions: capacity (how much data it can hold) and bandwidth (how fast data can move between memory and the compute units) For those interested in reading more of our work on the compute bottleneck, please reach out.

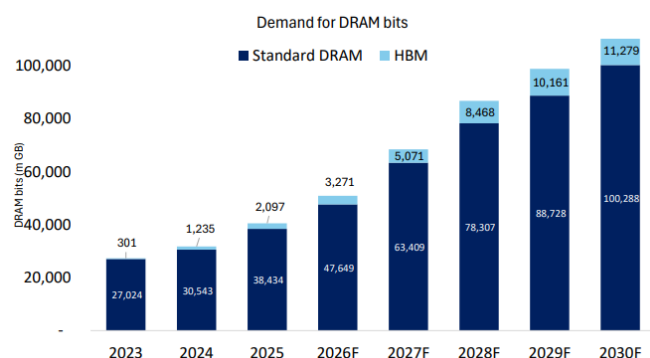
³³ https://news.futunn.com/en/post/74269829/micron-ceo-memory-is-the-overlooked-bottleneck-in-ai-supply?level=1&data_ticket=1781437411630811

⁴ [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

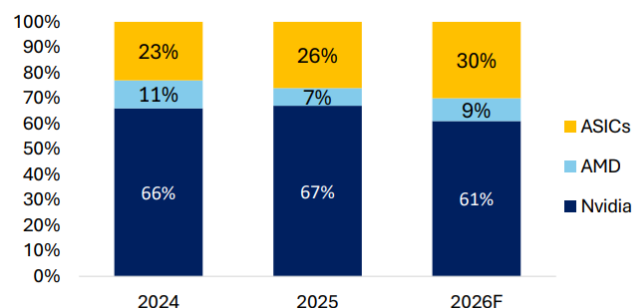


Our equity analysts forecast HBM demand to grow at a ~40% CAGR through 2030, versus a ~21% CAGR in demand for standard DRAM (the chips used for everyday consumer computing and standard cloud databases).⁵ This is because hyperscalers will pay premiums over automotive, consumer-electronics, and industrial buyers, and will sign multi-year lock-ins to secure supply.

Figure 2: Standard DRAM demand grows at +21% CAGR **Figure 3: HBM demand breakdown by major customers through 2030, vs HBM at +40%**



Source: Deutsche Bank equity analyst research.



Source: Deutsche Bank Equity Team. For more, see [here](#) and [here](#).

(ii) The supply side: Why fabs can't follow

The inability of supply to meet outsized demand is compounded by the lead-time needed to construct more fabs. Memory fabs (fabrication plants) are among the most expensive and complex facilities to build, with a 2-3 years construction timeline. Most announced projects will not contribute meaningfully to HBM capacity until 2027 at the earliest.

The complications of HBM production exacerbate the memory shortage crisis. HBM is made from DRAM, but producing one additional bit of HBM requires around 3x more silicon.⁶ That means every wafer directed to HBM consumes multiple wafers' worth of standard DRAM/NAND capacity needed for end-markets like automotives and PCs – creating shortage risk across not just the AI segment but all memory types. As this ratio moves from 3x to 4x silicon needed with HBM4/HBM4e, the crowding-out effect will worsen. For example, facilities and tools that would have served conventional memory are being redirected to HBM; “clean room space” – the ultra-sterile environments where wafers are processed – is at its limits, forcing manufacturers to choose what to build on finite factory floors. Anecdotally, Micron has said it can fulfil only 50% to 2/3 of demand from several key customers, describing the demand-supply gap as the largest it has ever seen.⁷ Qualcomm has also said that the size of the 2026 handset market will be defined by DRAM availability rather than consumer demand.

Our Deutsche Bank equity analysts' DRAM demand-supply model projects continually tight supply demand conditions in the coming years (beyond 2028). Growth in WSPM is expected to still lag accelerating demand for DRAM wafers over the coming years, even despite new fab announcements and project expansions by top DRAM suppliers.⁸

DRAM now represents ~70% of the memory market – up from its historical 50-60% range, as the HBM boom lifts DRAM's share of revenue.

⁵ [Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist](#)

⁶ Otherwise known as the HBM trade ratio.

⁷ <https://www.techradar.com/pro/we-are-only-able-to-supply-for-our-key-customers-in-the-midterm-about-50-percent-to-two-thirds-of-their-requirements-micron-ceo-forecasts-production-spend-increase-to-meet-the-insane-demand-for-memory-but-the-ram-crisis-will-only-get-worse>

⁸ WSPM stands for Wafer Starts Per Month, which measures the number of raw silicon wafers that enter a fab's manufacturing pipeline every month. That is the university industry standard metric to measure and express the max production capacity of a fab.



2. The spillover: Why it matters beyond AI

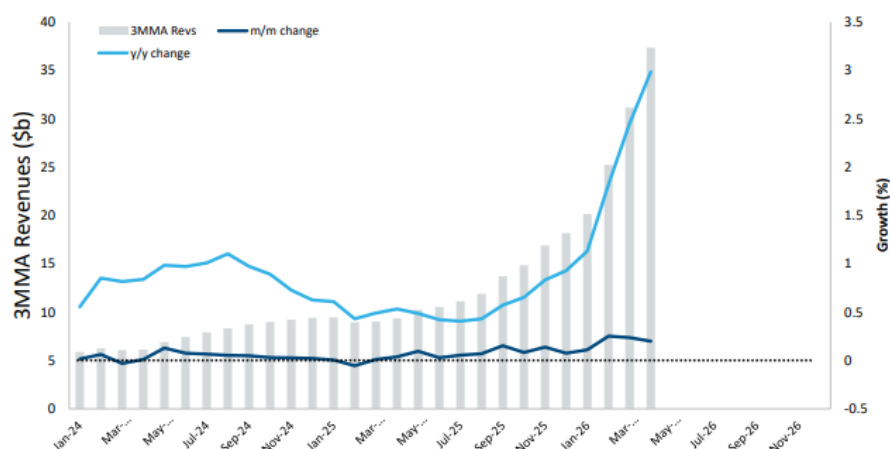
(i) Prices, mix shift, and the HBM-vs.-DRAM squeeze

The overwhelming demand for AI compute and inference means US hyperscalers will pay premiums to lock in future supply. Examples include SK Hynix's supply agreement with OpenAI for the Stargate Project, Samsung reported long-term deals with Google and Microsoft, and Micron's first five-year strategic agreement.

To meet surging AI demand, chip suppliers are reallocating their capacity towards more HBM production and pulling out of older segments. The result is an industry cycle shift, in which standard DRAM and NAND are also pulled into data centres and tighten supply, hitting non-HBM buyers in energy, consumer electronics and appliances. TrendForce predicts standard DRAM contract prices will rise 58-63% q/q in Q2 2026, with NAND flash contract prices up 70-75% q/q.⁹

We are seeing this today not just in leading-edge pricing: in Q1 2026, there was an uptick in lagging-edge pricing for the first time in two years, as increases filter into mature chip categories.¹⁰ Price rises are set to spread to server CPU, storage, substrates and passives, while several of the largest mature foundries – Vanguard, Powerchip and Nexchip – reported they would raise prices by up to 10% or more, driven by a 5% increase in costs.¹¹

Figure 4: 3MMA DRAM revenues continued to surge in April, growing to \$37.3bn (+299% y/y)



Source: Deutsche Bank equity analyst research, SIA. Notes: 3MMA = 3 month moving average.

(ii) Who can cope, and who gets rationed?

The shortage is affecting all direct users of memory. But hyperscalers and AI cloud providers remain the best positioned overall to weather it, because they can price AI compute services to reflect the underlying input costs.

With AI absorbing all the silicon, broader corporates and consumers are being squeezed and rationed. Lenovo, Dell Technologies, Apple and HP have sounded alarms on rising memory costs. The effect is global: Taiwanese manufacturers Acer and ASUS, and Chinese smartphone company Xiaomi have also issued warnings.

(iii) The most exposed sectors, bypass-through power

Of the key memory end-markets – data processing, communications, consumer and industrial – those most exposed to the AI squeeze are PCs (within data processing), consumer electronics (within consumer), and automotives (within industrials).

⁹ <https://www.trendforce.com/presscenter/news/20260331-12995.html>

¹⁰ Semis skimmed: Pricing and lead time tracker: Mar 2026. Deutsche Bank Equity Research.

¹¹ Semis skimmed: Pricing and lead time tracker: May 2026 - Deutsche Bank Equity Research



Starting with automotives: suppliers that sell complete technology systems to manufacturers such as Ford and Toyota have among the highest exposure to memory chips. These suppliers embed DRAM across the vehicles to help drivers use cockpit infotainment and navigation, advanced driver-assistance systems, and more. In 2026, companies such as Aptiv, Aumovio and Ford have signalled low DRAM supply despite efforts to secure inventory and contracts. With inventories set to drain through 2026, our equity analysts see potential impact on production volumes from 2027.¹²

Still, these suppliers may be best positioned among non-AI buyers, given their recovery mechanisms established after the 2021 chip shortages and these companies' ability to pass costs to the equipment manufacturers like Ford and Toyota, who have more limited consumer pricing power. Our DB models estimate the inflated cost of DRAM raises the price of an average vehicle by \$150-300 for luxury high-end models, and \$400-600 for vehicles with higher-level autonomy.¹³ These carmakers therefore can either absorb the cost and compress margins, pass it to consumers, or strip out some DRAM-intensive features such as L2+ autopilot capabilities or GenAI chatbots.

Consumer-electronics manufacturers and PCs are also highly exposed to the memory crisis, via margin compression and softer unit volume. Compared with automotives, this sector faces a greater challenge because it has the weakest pass-through capacity among the sectors. Price sensitive categories – smartphones, traditional phones, PCs, smart TVs, gaming consoles and tablets – all face rising memory costs per unit. Total consumer revenues also fell -10% q/q in Q1 this year, with most weakness from AMD's and Nvidia's consumer businesses. Our equity team expects total consumer end-market revenues down -15% y/y in 2026, recovering +9% y/y in 2027.¹⁴

The consequences of memory's supply disruption amongst consumer electronics are being echoed publicly across large corporations. In Apple's Q1 earnings call, its CEO warned that higher memory costs are just the beginning and that the company was facing supply constraints.¹⁵ While Apple has not detailed its options, it has reportedly quietly reduced the maximum memory capacity offered on several of its computers.¹⁶ It is not just Apple: Dell and Microsoft are also paring back their configurations to manage input costs, rather than pass them onto the individual buyer. or Microsoft, even as the company advances in AI with its Copilot+ – quietly cut its new surface for Business laptops to as little as 8GB of RAM (down from 16GB previously) on entry-level configurations to mitigate the DDR5 costs.¹⁷

Figure 5: Who is exposed?

Data Processing	Communications	Consumer	Industrials
PCs Servers	Wireless Wireline	Consumer electronics	Manufacturing Automotive Aerospace & Defence MedTech

Source: Deutsche Bank Research

¹² [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

¹³ [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

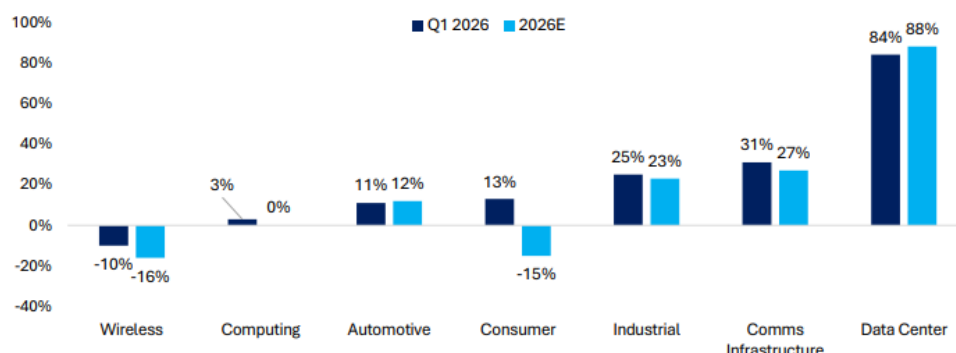
¹⁴ [US Semiconductors: Quarterly End-Market Monitor - Deutsche Bank Research](#)

¹⁵ [Apple CEO warns of memory crunch. 'We'll look at a range of options'](#)

¹⁶ [Apple quietly axes 128GB Mac Studio amid supply constraints and local AI frenzy — highest memory capacity reduced to 96GB, two months after discontinuation of 512GB model | Tom's Hardware](#)

¹⁷ <https://www.tomshardware.com/laptops/8gb-of-ram-is-back-on-laptops-companies-are-lowering-memory-offerings-to-make-affordable-notebooks-during-component-crisis>

Figure 6: End-market growth, Q1 vs. 2026 estimates



Source: Deutsche Bank equity analyst research

(iv) The macro spillover: the chipflation tax

With companies in consumer electronics and appliances for example forced to bear rising costs, the risk of a “memory inflation tax” on consumers is growing.

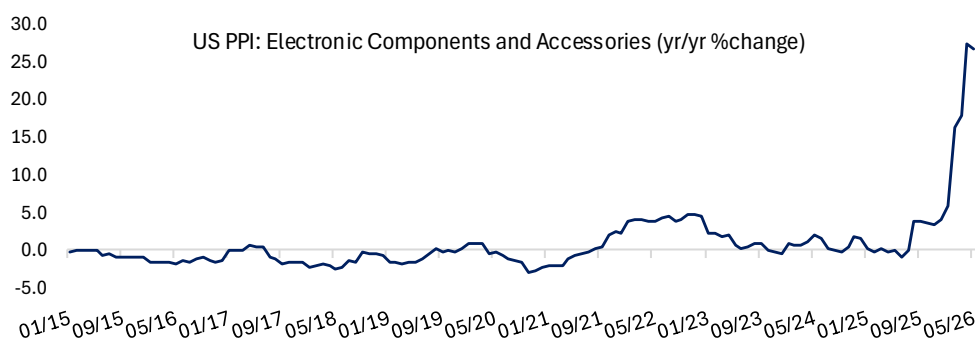
This is happening because end-product manufacturers – across PCs and smartphones for example – face severe Bill of Materials (BOM) cost pressure from the reallocation of silicon towards AI infrastructure. When memory prices surge, products with memory-heavy BOMs meet sharp production-cost increases, forcing companies to redesign or pass the “tax” down to consumers. We detailed the automotive and smartphone trade-offs above; if firms pass costs through, the average person takes a direct price impact. PC vendors Lenovo, Dell and ASUS have already warned of potential 15-20% price hikes and contract resets, with Lenovo and Dell starting increases this July.¹⁸

Consumers may also face financing and long-term stress: higher vehicle prices, for example, could push buyers into longer loan terms and higher total interest payments over the life of the asset.

In short, the memory-supply crisis has the potential to inflict acute second-order macroeconomic effects. The AI squeeze is crowding out the economy’s ability to serve the day-to-day needs of ordinary people. Qualcomm and AMD have both flagged memory constraints actively hitting consumer-device demand.

We are already seeing it in US purchasing price inflation (PPI): the electronic components and accessories – which captures semiconductor and memory-devices prices – rose +26.9% y/y in May, up from +5.9% y/y rise in January. That, in turn, is pushing up upstream input costs between the businesses that trade these materials.

Figure 7: The sharp rise in PPI for electronics is driven by surging memory-driven chip costs



Source: Deutsche Bank Research, Haver.

¹⁸ <https://www.trendforce.com/news/2026/06/10/news-lenovo-reportedly-set-for-july-price-hikes-across-product-portfolio-as-memory-costs-pressure-pc-market/>



3. Fault lines: The geopolitics of memory

(i) Winners and losers, country by country

South Korea is heavily exposed to the energy-intensive global AI capex boom, as its two tech giants (SK Hynix and Samsung) benefit from specializing in memory for AI data centres. The two companies account for around 69% of global DRAM production. However, that success heavily dictates Korea's economy, and when memory falls it drags Korean markets down with it. For example, after semiconductors melted down on June 8, the tech-heavy Kospi fell -8.29%, its ninth worst day in more than 12,000 trading sessions since Bloomberg data began in 1980. And earlier, in late March, Google's publication of a research algorithm called TurboQuant – which reduces the memory needed to run AI inference – also sent SK Hynix and Samsung shares lower. In short, the winners of the AI trade can quickly become the biggest losers if the gamble for global AI buildout fails.

China's semiconductor exports have also risen, as firms like CXMT (which specializes in DRAM), and Yangtze Memory Technologies (which specializes in NAND) capitalize on industry's shift towards HBMs, which has left supply for standard DRAMs tighter. Exports surged by 72.6% y/y (in dollar terms) between Jan-Feb 2026, with memory the largest and fastest-growing category: customs data show memory exports soared 81.6% y/y (in dollar value) in Dec 2025.¹⁹

At the same time, Chinese companies are also hurt by the memory squeeze, as waiting lists for advanced memory and compute lengthen. Vendors (CXMT, JHICC) are expanding aggressively but are constrained to lagging-edge DRAM by US export controls. For example, CXMT's Shanghai fab expansion will add up to 200k wafers per month by 2030 but only begins high-volume manufacturing in 2027. Chinese producers will relieve some pressure in low-end commodity segments but cannot address the HBM or automotive-grade DRAM shortage. US export controls have severely limited domestic fabrication capacity, so Chinese supply cannot substitute for the premium segments most affected. Ultimately, Chinese memory companies will be confined to lower-performance end-uses and will remain well behind in HBM.

The United States is home to Micron (24% of global market share) and is the largest region by memory revenue. Outside AI buyers, there is real potential for US corporates and government to lean on China's lagging-edge advantage as a partial answer to the AI memory squeeze. For instance, in February, the Pentagon quietly removed China's CXMT and YMTC from its restricted-companies list, which makes US sourcing Chinese memory and NAND more plausible – though it is too early to judge whether adoption follows.²⁰ Several tech companies, including Apple and major PC OEMs like HP and Dell, are reportedly exploring or qualifying partnerships with Chinese memory makers to diversify supply chains and combat rising memory costs.

The geopolitical stakes are serious. Every dollar into CXMT and YMTC also benefits companies widely viewed as national-security risks – a hard case to make when the current US administration's priority has been to decouple from China (as evident from the MATCH Act). Whether US tech companies actually embark on these partnerships is uncertain. As noted earlier, China is confined to the lagging edge, so it would be difficult to justify a company like Apple shipping more phones built on a less advanced node than prior versions.

Congress has introduced the US MATCH Act, which aims to restrict countries from buying semiconductor manufacturing equipment technology from the US and its trade partners that they cannot build themselves. Importantly, MATCH designates restrictions on CXMT and YMTC across exports, servicing, and technical support. It has received broad bipartisan support but blowback from US chipmakers, who warn of further memory price

¹⁹ Gavekal, China Customs Administration

²⁰ https://wccftech.com/cxmt-ymtc-removed-from-pentagon-list-opening-door-for-chinese-dram-adoption/?utm_source=substack&utm_medium=email



risers, as the Act could cap global production capacity and shift pricing power back to a dominant Western-aligned oligopoly. Micron alone has publicly backed MATCH – unsurprisingly given it is in the company’s interest to maintain price advantage.²¹ If MATCH were to be implemented, it would have severe consequences for China’s ability to industrialize.²²

4. Breaking the bottleneck: What could actually fix it

(i) Building more fabs (and why it’s slow)

To meet ongoing demand, the big 3 players are raising capex to help break the memory bottleneck. Our latest DB estimates place total DRAM wafers per month estimate to increase by +1,475k WSPM over the next five years, due to a variety of new fab announcements/expansions/accelerations.²³ Micron remains at the heart of a transformative moment in the memory industry, with a \$200bn commitment to expand its memory capacity in the US, and will benefit from sustained tightness in demand. As we expressed earlier, however, fab construction is a multi-year process and will come online only after the peak of the shortage in the industry occurs.

Samsung, SK Hynix and Micron are trying to pull the timeline forward by buying partially finished or existing facilities. One example is Micron’s \$1.8bn acquisition of a used fab from Powerchip Semiconductor Manufacturing Corporation (PSMC) in Taiwan earlier this year, which lets Micron skip the roughly 2-year timeline needed to build a facility from scratch.²⁴ We expect other memory companies to follow in Micron’s lead. Samsung is also reportedly negotiating long-term deals with Google and Microsoft, while Micron has disclosed its first five-year strategic customer agreement.

Whether China can be a credible player is still hard to gauge, as many new Chinese fabs remain under construction. But as we said before, these would primarily address lagging-edge memory demand, and the US export controls have severely limited China’s ability to fabricate at more advanced nodes.

(ii) Beyond fabs: Algorithms, SRAM and yield

Google’s unveiling in March of an algorithm that requires less memory to serve LLMs (“TurboQuant”) caused a sharp decline in the share price of Samsung (-6%), Micron (-7%) and SK Hynix (-7%) on the day.

However, the subsequent recovery in the memory market underscores that experiments like TurboQuant have yet to convince investors or AI accelerators that they a breakthrough is at hand that would turn the memory boom to bust. The key reason is scope: TurboQuant compress the inference-time KV cache – not model weights or training memory. While both focus on reducing memory bottlenecks, KV caching speeds up generative inference, whereas training memory optimization allows models to process massive datasets during the learning phase. Therefore, TurboQuant’s potential impact was only limited to a part of total AI memory demand. More fundamentally, because cheaper inference historically expands total usage – the Jevons Paradox – such efficiency gains may ultimately lift, not cut, aggregate memory demand.

Still, TurboQuant is likely the first of several attempts to reduce reliance on HBM. Other strategic efforts include improved KV-cache eviction (such as Nvidia’s inference context memory storage system) and greater reliance on SRAM.

There will likely be more attempts to upgrade node conversion (upgrading the manufacturing process to print smaller circuits) and yield improvements (the share of chips on a wafer that actually work). Both aim to lift the capacity of existing memory

²¹ <https://finance.yahoo.com/economy/policy/articles/micron-backs-match-act-policy-180733909.html>

²² https://www.kim.senate.gov/press_release/senators-kim-and-ricketts-introduce-match-act-to-level-the-global-playing-field-for-u-s-tech/

²³ [Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist - Deutsche Bank Research](#)

²⁴ <https://investors.micron.com/news-releases/news-release-details/micron-signs-letter-intent-purchase-tongluo-site-begin-0>



production rather than building new fabs. The memory industry will nonetheless continue to face headwinds in bringing incremental capacity online.

Conclusion: From commodity to macro variable

The production of memory chips is becoming a zero-sum game. For every wafer devoted to HBM stacks for AI servers, others are unavailable for smartphones, PCs, or vehicles. And because the shortage will reverberate across key economic systems well beyond AI – hurting the average consumer via memory-cost inflation – memory chips have transitioned from a pure commodity to a distinctly macroeconomic variable.

The companies best positioned to survive the structural dynamics of the memory industry are, of course, the hyperscalers and AI cloud providers. And Micron stand out as a clear winner of the current shortage. For traditional PCs and smartphone manufacturers, memory is critical to device function, but unlike in AI, adds little value at higher performance or capacity; their objective is to minimize cost per bit while keeping systems efficient. As a result, the insatiable needs of the AI buildout are consuming capacity that would have gone to consumer-end markets. Consumer electronics, automobiles and automotive-system suppliers will face headwinds large enough that sustained DRAM inflation will have a measurable inflationary impact in 2027, hurting average customers.

Looking ahead, the question worth considering is what the macroeconomic consequences would be if the AI gamble fails. Where would all the HBM and silicon capacity go? The sharp swings triggered by events like TurboQuant show that if AI demand moderates even modestly, the market could swing from severe shortage to meaningful oversupply relatively quickly.

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